

ADV-DEVOPS EXPERIMENT-10

Aim: To install and deploy Nagios.

Theory:

1. Introduction to Nagios:

Nagios is an open-source monitoring system that enables organizations to identify and resolve IT infrastructure problems before they affect critical business processes. It is widely used to monitor system metrics, network protocols, applications, servers, and services. Nagios provides alerts and notifications when issues arise and can automatically resolve problems using event handlers.

2. Purpose of Nagios Installation and Deployment:

The main objective of installing and deploying Nagios is to establish a centralized monitoring system that ensures the health, performance, and availability of IT infrastructure. It helps in:

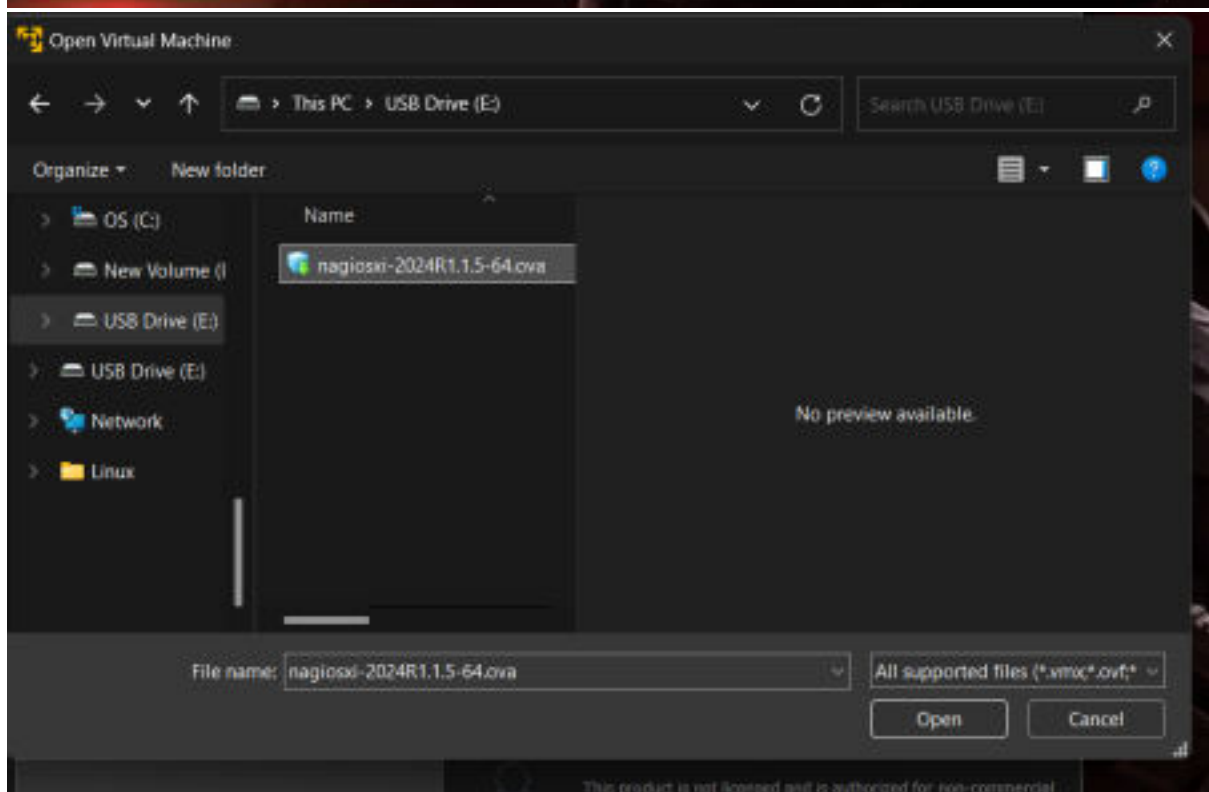
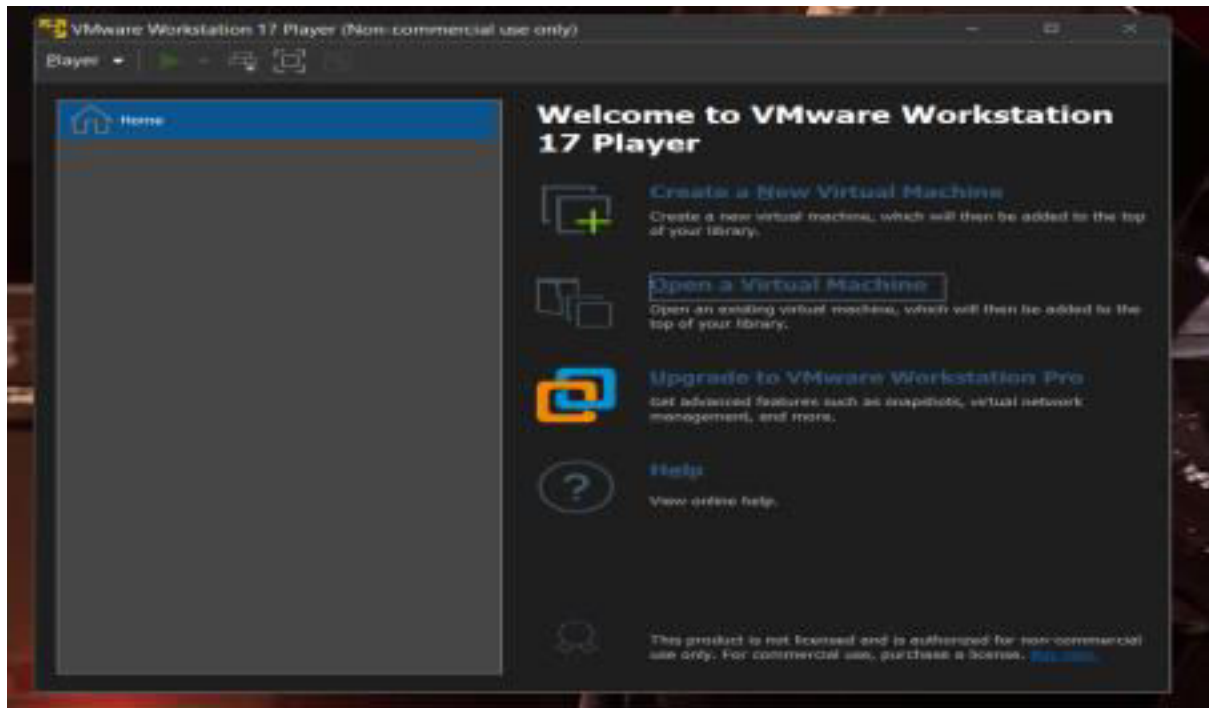
- Monitoring network services (SMTP, POP3, HTTP, SNMP, etc.)
- Monitoring host resources (CPU load, disk usage, memory, etc.)
- Notifying users when service or host problems occur
- Escalating alerts to appropriate personnel
- Visualizing the status and history of hosts and services

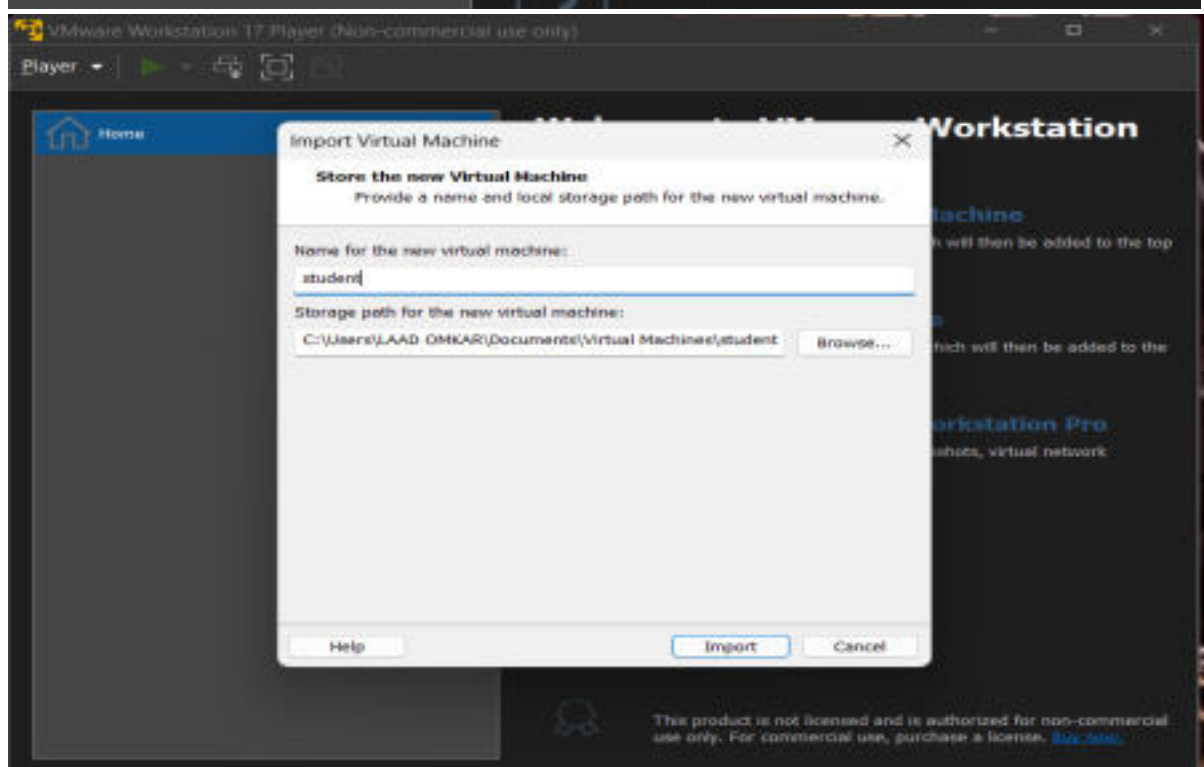
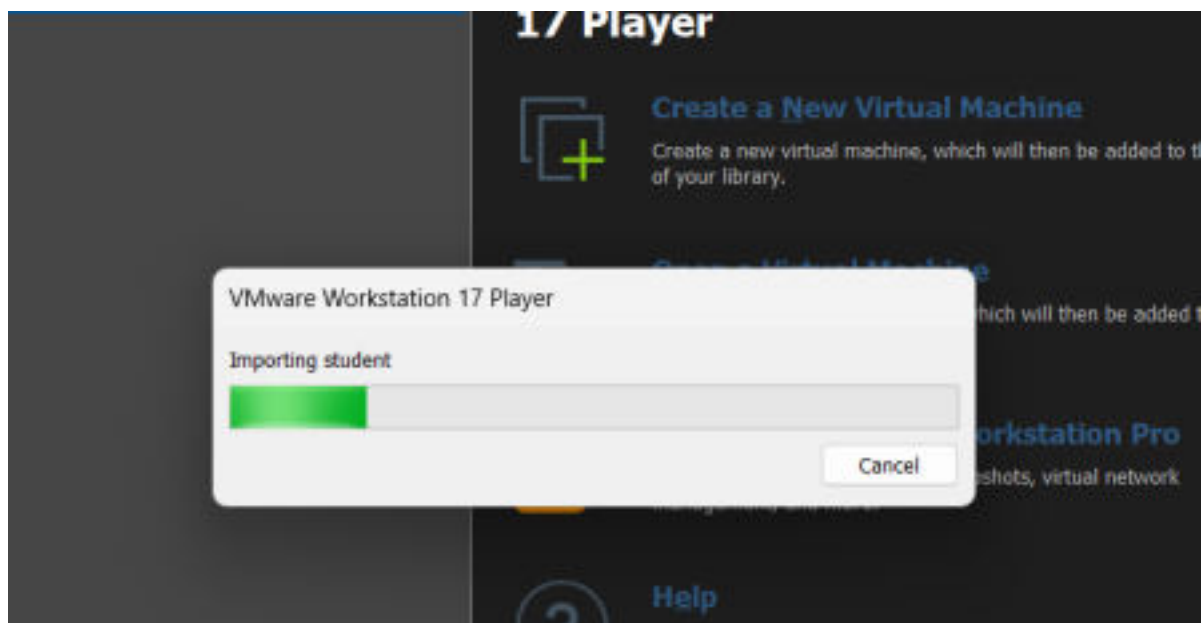
3. Components of Nagios

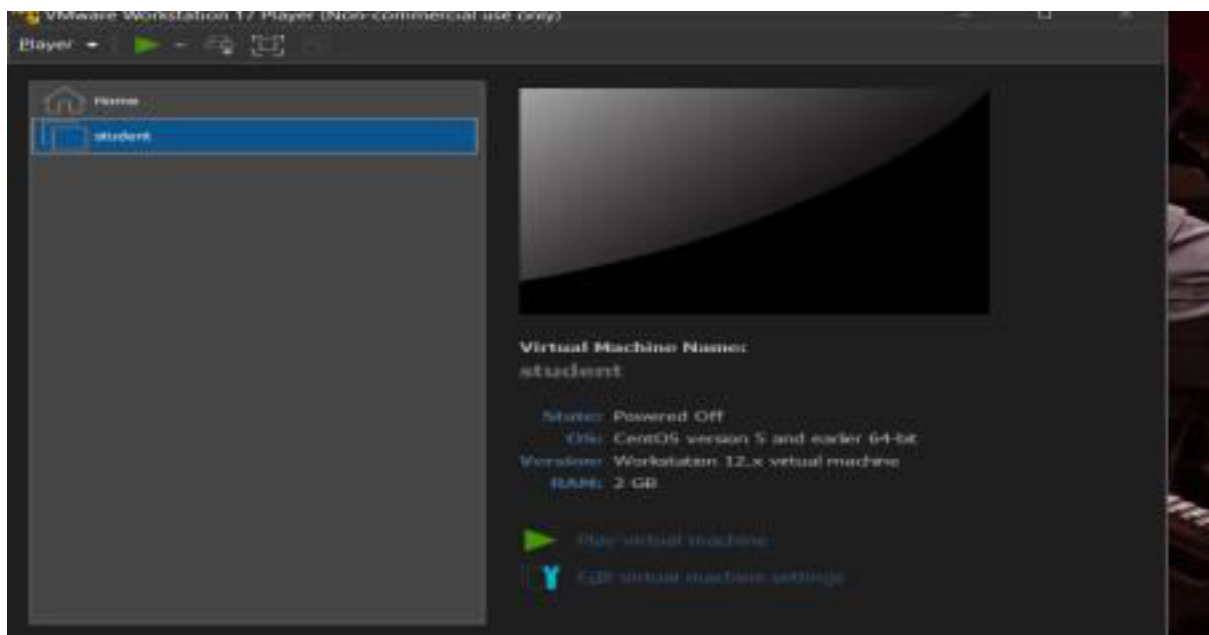
The main components of a Nagios deployment include:

- **Nagios Core:** The central engine that schedules checks and handles alerts.
- **Plugins:** Small executables that perform the actual checks.
- **NRPE (Nagios Remote Plugin Executor):** Allows monitoring of remote Linux/Unix machines.
- **Web Interface:** Allows users to view monitoring status, logs, and reports.
- **Configuration Files:** Define hosts, services, commands, contacts, and notification rules.

Procedure:





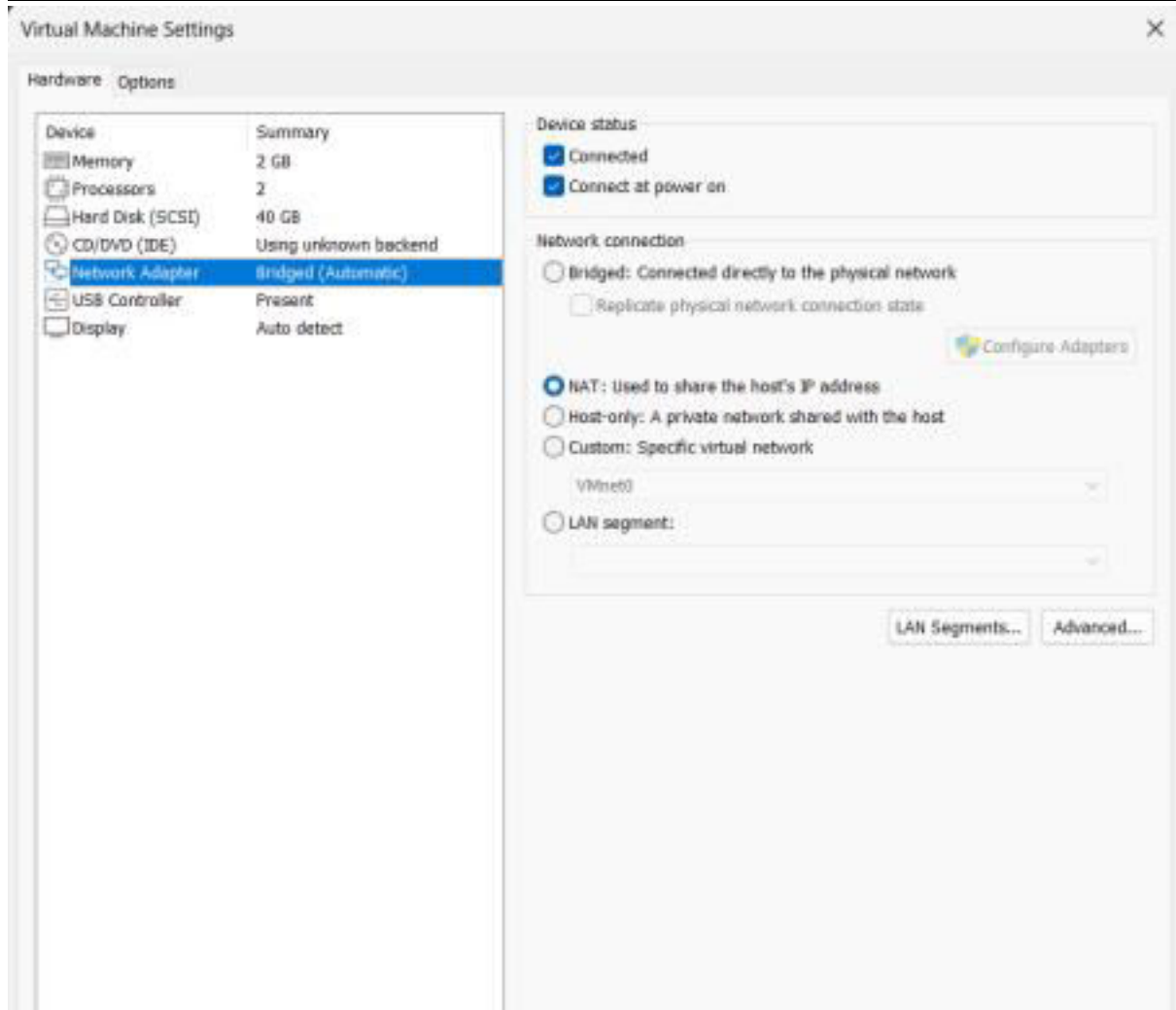


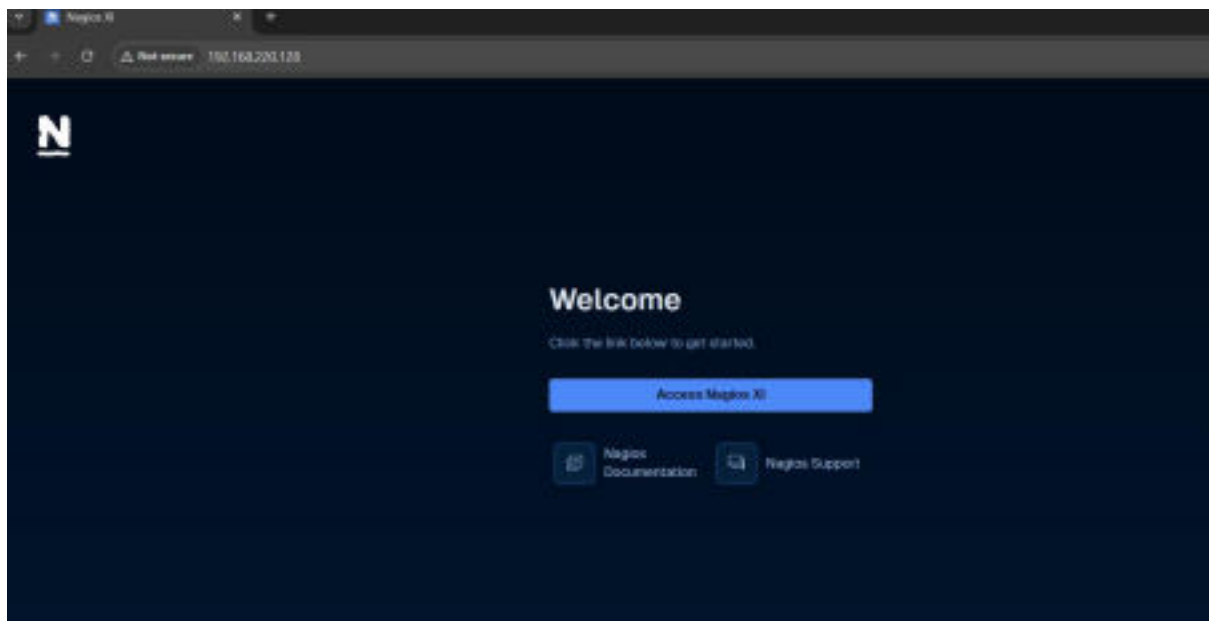
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[OK] 1 Reached target Network.
       Starting Network Manager Unit Online...
       Starting The Apache HTTP Server...
       Starting MySQL 5.6 database server...
[OK] 1 Started Syslog Process Control Daemon.
       Starting The PHP FastCGI Process Manager...
       Starting Postfix Mail Transport Agent...
       Starting OpenSSH server daemon...
       Starting Permit User Sessions...
[OK] 1 Started Network Manager Script Dispatcher Service.
       Finished Permit User Sessions.
[OK] 1 Started Command Scheduler.
[OK] 1 Started OpenSSH server daemon.
[OK] 1 Started Postfix Mail Transport Agent.
[OK] 1 Started The Apache HTTP Server.
       Starting Authorisation Manager...
[OK] 1 Started The PHP FastCGI Process Manager.
[OK] 1 Started Authorisation Manager.
       Starting Network Manager Script Dispatcher Service...
[OK] 1 Started Network Manager Script Dispatcher Service.
       Finished Network Manager Unit Online.
[OK] 1 Reached target Multi-User System.
       Starting Crash recovery kernel auditing...
       Starting netconsole.service: local Compatibility...
       Starting System Logging Service...
       Starting Simple Network Management Protocol (SNMP) Trap Daemon...
[OK] 1 Started Syslog startup script to update netconsole with proper network address and info.
[OK] 1 Started System Logging Service.
[OK] 1 Started Simple Network Management Protocol (SNMP) Trap Daemon.
       Starting SNMP Trap Translator (SNMPTT)...
[OK] 1 Started SNMP Trap Translator (SNMPTT).
[OK] 43940322 upload-journal@06051: Time jumped backwards, rotating.
[OK] 1 Created slice User Slice of UID 1000.
       Starting User Session Directory /run/user/1000...
[OK] 2 Finished User Session Directory /run/user/1000.
       Starting User Manager for UID 1000...
[OK] 1 Started User Manager for UID 1000.
[OK] 1 Started Session 1 of user session.
[OK] 1 Started Session 10 of user session.
[OK] 1 Started Session 11 of user session.
[OK] 1 Started Session 2 of user session.
[OK] 1 Started Session 3 of user session.
[OK] 1 Started Session 4 of user session.
[OK] 1 Started Session 5 of user session.
[OK] 1 Started Session 6 of user session.
[OK] 1 Started Session 7 of user session.
[OK] 1 Started Session 8 of user session.
[OK] 1 Started Session 9 of user session.
[OK] 1 (1 of 3) A start job is running for netconsole.service: local Compatibility (53s / no limit)

```







Nagios XI Installation

Finalize your Nagios XI installation and step the initial configuration. These settings can be changed later.

General System Settings

Program URL

`http://192.168.220.128/nagiosxi/`

The default URL, that is used to access Nagios XI on this server directly from your internal network.

Timezone

(UTC+05:30) Kolkata

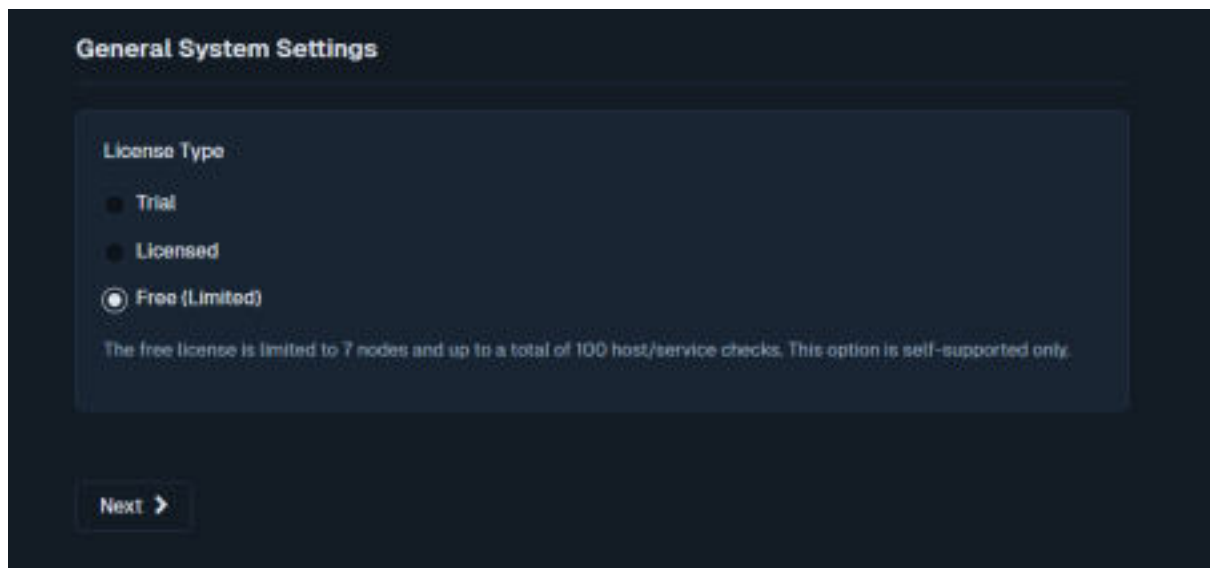
Language

English (English)

User Interface Theme

Neptune

☐ Use HTTPS only (all HTTP requests will be redirected to HTTPS) ⓘ

**Conclusion:**

The installation and deployment of Nagios serve as a critical step in establishing a robust IT monitoring environment. By implementing Nagios, organizations gain real-time visibility into the performance and availability of their network, servers, and services. Its modular architecture, plugin support, and flexible configuration make it a powerful and scalable solution for proactive infrastructure management. Proper deployment ensures timely alerts, minimizes downtime, and enhances overall operational efficiency. Thus, mastering the installation and deployment process of Nagios is essential for any system administrator aiming to maintain a reliable and secure IT environment.